Progressive exercises

E1 · Direct application. After determining the set on which it is defined, resolve Equation ln $(2x-1) = \ln(x+3)$.

E2 · Application and justification. Determining the real x for which $\ln x + \ln(x-2)$ is defined, then resolve ln $x + \ln(x-2) = \ln 3$.

E3 · Method to be organised. It is considered $f(x) = \ln(x^2+1)$ and $g(x) = x\ln x$. Clarify their set and to calculate their derivative.

E4 · Choice of strategy. Determine $\lim\limits_{x\to+\infty} \dfrac{\ln x}{\sqrt{x}}$ and $\lim\limits_{x\to 0^+} x^2 \ln x$. Justifying growth results comparative employees.

E5 · Synthesis or demonstration. Using the study of a function, demonstrate that $\ln x \leq x-1$ for all real $x > 0$. The concavity of the logarithm shall not be used directly.